J1
+12V
GNDA
GNDA
−12V
BUS +12V/G/G/−12
J2
+5V
GNDD
+5V/RTN (mod.1 J6)
PWR_FLAG
PWR_FLAG
PWR_FLAG
PWR_FLAG
PWR_FLAG
PWR_FLAG
C1 +12V 100n GNDA
C2 −12V 100n GNDA
C3 +5V 10u GNDA
C4 +3V3 100n GNDA
U1 AD5686RBRUZ
+5V +3V3
SCLK SDIN SYNC LDAC RESET RSTSEL GAIN
VDD VLOGIC VOUTA VOUTB VOUTC VOUTD VREF SDO GND
DAC_A DAC_B DAC_C VREF
GNDA GNDA DAC_RESET
C5 VREF 100n GNDA
U3 RPi 3B+ GPIO
+3V3 GNDD BTN_UP LCD_D7 GNDD LCD_D6 BTN_DOWN LCD_D5 GATE_IO GNDD LCD_D4 LCD_EN SDIN GNDD SCLK LCD_RS GNDD SPI_CE1 DAC_RESET GNDD GNDD BTN_BACK BTN_ENTER GNDD
U2A TL074H R1 10k R2 1k R7 DAC_A CV1 GNDA
U2B TL074H R3 10k R4 1k R8 DAC_B CV2 GNDA
U2C TL074H R5 10k R6 18k R9 DAC_C CV3 GNDA
J4
CV1 GNDA
CV2 GNDA
CV3 GNDA
GATE_OUT GNDA
GATE_OUT
J8 44780 10x2
GNDD +5V LCD_VO LCD_RS GNDD LCD_EN LCD_D4 LCD_D5 LCD_D6 LCD_D7 LCD_BLA GNDD
RV1 +5V LCD_VO GNDD 10k
+5V LCD_BLA 47R
R11 +12V GATE_INV 10k
Q1 2N7002 GATE_IO GATE_G GATE_INV
R14 1k GATE_G 10k GNDD
R12 +12V GATE_12 2k
Q2 2N7002 GATE_INV GATE_12
R13 GATE_12 10k GNDD
U2E TL074H +12V −12V
U2D TL074H GATE_12 GATE_OUT R15 1k
SW1 UP BTN_UP GNDD R16 10k +3V3 BTN_UP
SW2 DOWN BTN_DOWN GNDD R17 10k +3V3 BTN_DOWN
SW3 BACK BTN_BACK GNDD R18 10k +3V3 BTN_BACK
SW4 ENTER BTN_ENTER GNDD R19 10k +3V3 BTN_ENTER

GATE con dos MOSFET para que el reposo sea BAJO al arrancar
CV 0–7 V (la tesis usa 3–10). Presupuesto en ../tuning_budget.py
VLOGIC del DAC a +3V3, no a +5V: si no, VIH=3.5V y el Pi no llega
GNDA (bus, J1) y GNDD (retorno dedicado, J2) NO se unen en esta placa
SAM17 — Sintetizador analogico modular

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File: sam17-midicv.kicad_sch

Title: Modulo 02 — MIDI/CV: DAC 16 bits, 3x CV 1V/oct, GATE

Size: A3	Date: 2026-08-15	**Rev: A**
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